

Program Question_3

```
Implicit none
Real::x,res
!Value to Interpolated
x=38.0
Call BackwardDifference(x,res)
Write(*,*)"The Interpolated value at x=",x,"is",res
Contains
Subroutine BackwardDifference(x,res)
  Implicit none
  Real,Intent(In)::x
  Real,Intent(out)::res
  Real,Dimension(6)::angle,sines
  Real,Dimension(6,6)::delta
  Real::h,u
  Integer::n,i,j

  !Intialize the valuse
  angle=[15.0,20.0,25.0,30.0,35.0,40.0]
  sines=[0.2588190,0.3420201,0.4226183,0.5,0.5735764,0.6427876]
  n=6

  delta(:,1)=sines
  Do j=2,n
    Do i=n,j,-1
      delta(i,j)=delta(i,j-1)-delta(i-1,j-1)

    End Do
  End Do
  !stepsize
  h=angle(2)-angle(1)
  u=(x-angle(n))/h
  res=sines(n)
  Do j=2,n
    res=res+(u*delta(n,j))/(factorial(j-1))
    u=u*(u+j-1)
  End Do
End subroutine
!For Factorial
Integer Function factorial(n)
Implicit none
Integer,Intent(In)::n
Integer::i
factorial=1
Do i=1,n
  factorial=factorial*i
End Do
End function
```

End Program